



TEST CODE **02220020**

FORM TP 2004199

MAY/JUNE 2004

CARIBBEAN EXAMINATIONS COUNCIL

ADVANCED PROFICIENCY EXAMINATION

ENVIRONMENTAL SCIENCE

UNIT 2: AGRICULTURE, ENERGY AND ENVIRONMENTAL POLLUTION

PAPER 02

2 hours 30 minutes

10 MAY 2004 (p.m.)

READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. **DO NOT** open this examination paper until instructed to do so.
2. This paper consists of **NINE** questions, **THREE** from each Module.
3. Answer **ANY TWO** questions from **EACH** Module; a total of **SIX** questions.
4. Write your answers in the answer booklet provided.
5. The use of a non-programmable calculator is allowed.

MODULE 1

Answer ANY TWO questions.

1. In an experiment to monitor the impact of pesticides on agricultural pests and their predators, the populations were monitored over a period of 110 days after which a pesticide was applied to the field. Figure 1 shows the variation in the populations of the pest and its natural predator before and after the application of the pesticide.

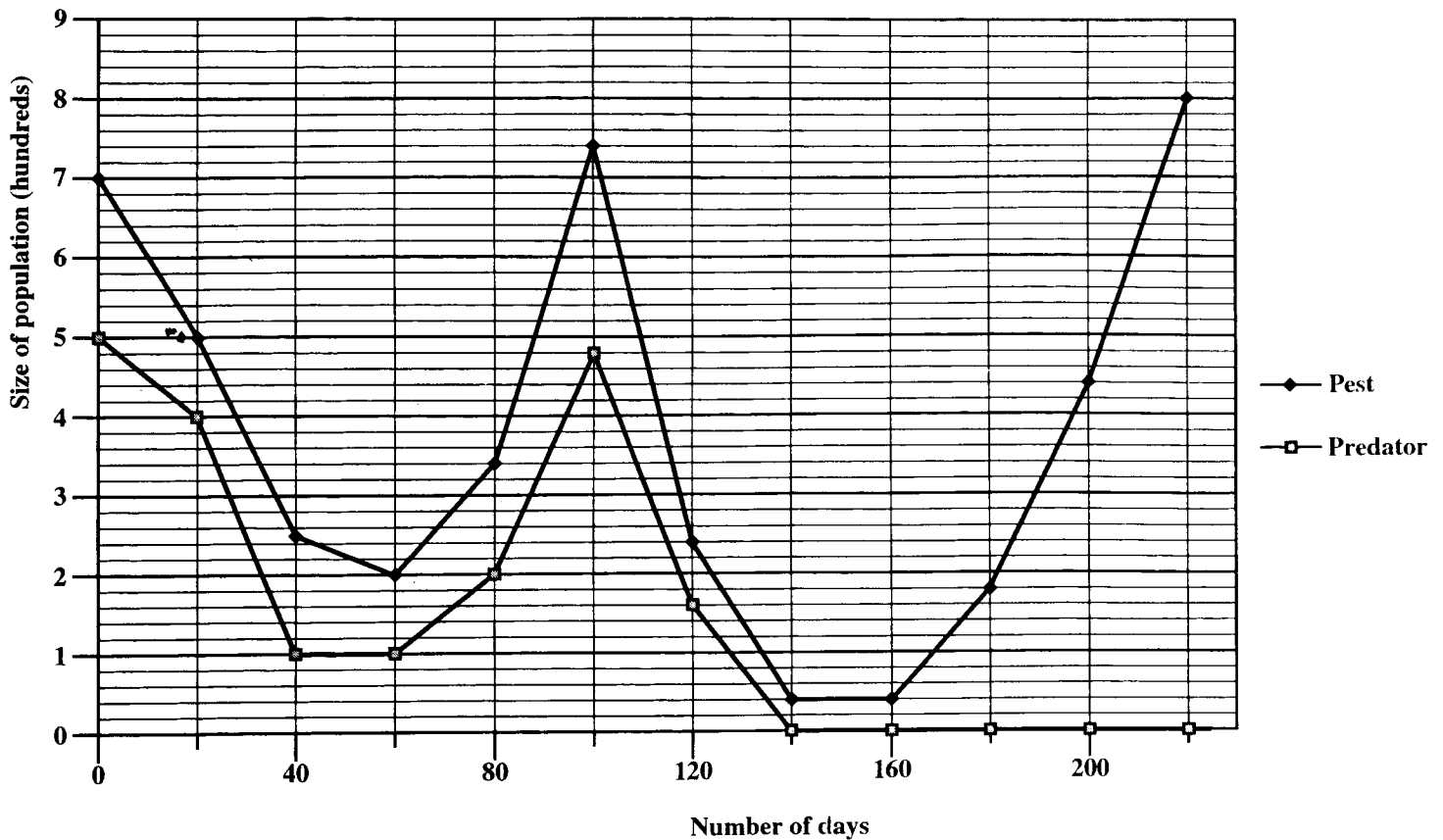


Figure 1. Variation of populations

- (a) Describe the importance of pesticide use in agriculture in the Caribbean. [4 marks]
- (b) Explain the observations after the application of the pesticide as shown in Figure 1. [6 marks]
- (c) Describe ONE similarity and FOUR differences between the use of biological and chemical methods to control agricultural pests in the Caribbean. [10 marks]

Total 20 marks

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2. “Despite the dominant role of agriculture in the overall socio-economic development of the Caribbean, a number of related socio-environmental issues severely threaten the sustainability of agricultural productivity in the Region”.

By discussing EACH of the following, support the statement above.

- (a) Health issues [6 marks]
- (b) Threats to the sustainable livelihood of communities [6 marks]
- (c) Land availability [8 marks]

Total 20 marks

3. Evaluate EACH of the following as environmentally sustainable practices in agricultural systems of the Caribbean.

- (a) No tillage farming [5 marks]
- (b) Contour farming [5 marks]
- (c) Organic farming [5 marks]
- (d) Crop rotation [5 marks]

Total 20 marks

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MODULE 2

Answer ANY TWO questions.

4. Table 1 below shows the electricity generation by source from 1994 to 1999 for a Caribbean country.

TABLE 1: ELECTRICITY GENERATION BY SOURCE, 1994 – 1999

YEAR	Sources (000 Megawatthours)	
	Oil Generators	Hydroelectric Generators
1994	1475.5	114.5
1995	1550.9	111.1
1996	1585.0	109.8
1997	1637.2	99.8
1998	2041.2	89.7
1999	2078.3	86.8

- (a) From the data presented in Table 1, outline the trends in energy generation in this country. [4 marks]
- (b) Outline FOUR reasons, why this trend of energy use identified in 4 (a) is undesirable. [8 marks]
- (c) (i) Discuss ONE way that the country can diversify its source of energy. [4 marks]
- (ii) State TWO advantages and TWO disadvantages of the approach discussed in (c) (i). [4 marks]

Total 20 marks

5. Currently, some Caribbean countries are considering the use of liquid natural gas (LNG) as a way of decreasing their dependence on oil to generate electricity.

- (a) What is 'liquid natural gas (LNG)' ? [2 marks]
- (b) Assess the use of LNG as a source of energy for generating electricity with reference to its production, transportation and use. [12 marks]
- (c) Comment on the appropriateness of the course of action being considered by the Caribbean countries mentioned above. [6 marks]

Total 20 marks

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6. (a) Using ONE example in EACH case, distinguish between 'energy efficiency' and 'energy conservation'. [6 marks]
- (b) (i) Explain what is meant by 'control of demand patterns' or 'demand management'. [2 marks]
- (ii) Suggest FOUR ways in which electricity generating companies in the Caribbean can achieve demand management. [4 marks]
- (c) Discuss TWO advantages and TWO disadvantages of implementing demand management initiatives at electricity generating companies in Caribbean countries. [8 marks]

Total 20 marks

MODULE 3

Answer ANY TWO questions.

7. (a) Explain the importance of the water quality parameters, BOD, TSS and TN, in monitoring water pollution. [6 marks]
- (b) Identify ONE possible cause and ONE source in Caribbean countries that can influence EACH of the parameters in (a) above. [6 marks]
- (c) (i) Describe TWO environmental impacts associated with water pollution. [6 marks]
- (ii) For EACH impact identified, suggest ONE method to mitigate it. [2 marks]

Total 20 marks

8. (a) (i) Explain what is meant by 'the greenhouse effect'. [4 marks]
- (ii) Explain the importance of the greenhouse effect to life on earth. [2 marks]
- (b) Discuss TWO factors which have contributed to global warming and the extent to which Caribbean countries have contributed to this phenomenon. [8 marks]
- (c) Describe TWO possible effects of global warming that may affect Caribbean countries. [6 marks]

Total 20 marks

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9. Most solid waste disposal sites in the Caribbean are dumps. The pie chart in Figure 2 shows the typical percentage composition of solid waste arriving at dump sites.

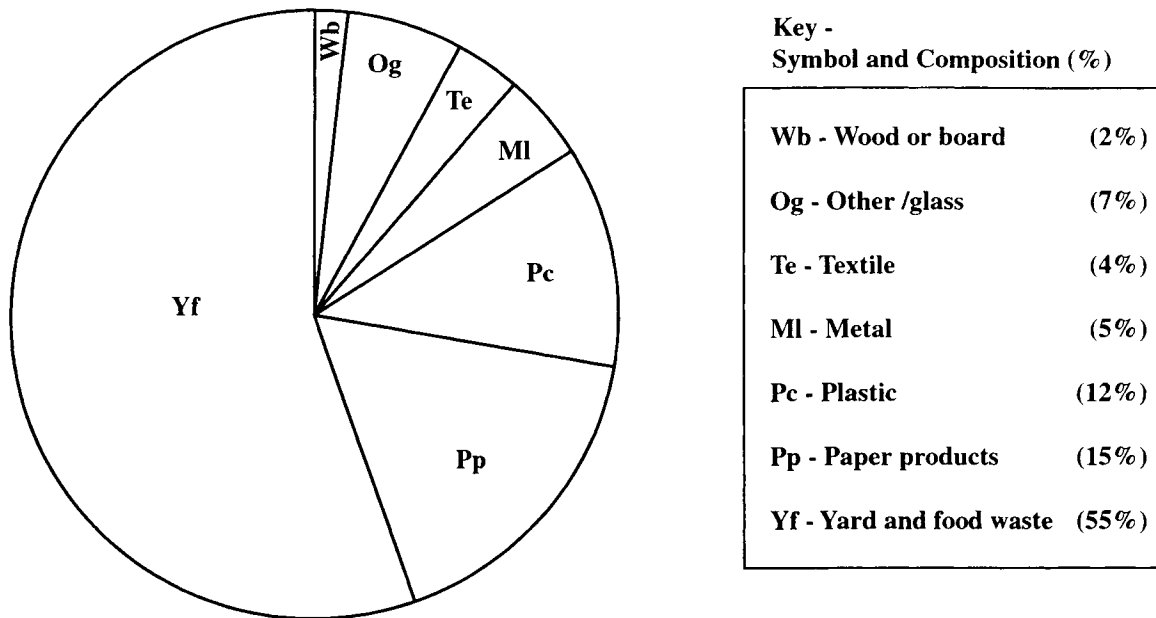


Figure 2. Composition of solid waste

- (a) Outline TWO problems associated with the use of dump sites in the Caribbean.
[4 marks]
- (b) Based on the composition of solid waste shown in Figure 2, discuss how the volume of the solid waste that reaches the dump could be reduced.
[8 marks]
- (c) Discuss an alternative disposal method that is appropriate for Caribbean countries.
In your discussion, include THREE advantages and ONE disadvantage of this method.
[8 marks]

Total 20 marks

END OF TEST